



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II – JANUARY, 2020
ST102/MT102 – INTRODUCTION TO PROBABILITY THEORY

Time Allowed: **TWO Hours**

Answer **All** Questions.

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1. a). If eight people are to be seated in a row, calculate the following probabilities.
- i) P(Two persons A and B always sit together)
 - ii) P(If five are men, they always sit next to each other)
 - iii) P(If four are men and four are women, no two men or two women sit next to each other)
 - iv) P(If they are four married couples and each couple must sit together)
- b). If coin A is flipped, it comes up heads with probability $1/4$. If coin B is flipped, it comes up heads with probability $3/4$. Suppose that one of the coins is randomly selected and is flipped twice. If both flips land heads, what is the probability that coin B was the one flipped?
2. The probability density function of X , the life time of a certain type of electronic component (measured in hours), is given by
- $$f(x) = \begin{cases} \frac{10}{x^2} & ; x > 10 \\ 0 & ; \leq 10 \end{cases}$$
- a). Show that $f(x)$ is a legitimate pdf.
 - b). Find the expected life time of these components.
 - c). Derive the Cumulative Density Function (CDF) of X .
 - d). Calculate $P(X > 20)$.
 - e). What is the probability that of 6 such types of devices, at least 3 will function for at least 15 hours? What assumptions are you making?

[P.T.O.]

3. a). Suppose X has a Geometric (p) distribution. Show that the moment generating function of X is

$$M_X(t) = \frac{pe^t}{1 - (1-p)e^t} \quad \text{for } t < -\ln(1-p)$$

- b). A player will win if he can get 1 or 6 as the outcome of rolling a balance six sided die.
- (i) If he get ten chances to roll the die, what is the probability that he will win exactly 5 times?
 - (ii) What is the probability that his first win occurs at the 6th roll?
 - (iii) What is the probability that his 3rd win occurs in the 5th roll?
 - (iv) Approximate the probability of having 15 or more wins if he get the chance to roll the die 60 times. Clearly state the assumptions you make.

4.

Let a continuous random variable X follows uniform distribution within the interval $[\alpha, \beta]$.

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & ; \alpha \leq x \leq \beta \\ 0 & ; \text{o/w} \end{cases}$$

- a). Derive an expression for the k^{th} moment about the origin.
- b). Use the answer in part (i) to calculate the mean and the variance of X .
- c). Find the lower bound for the probability that X is 2 standard deviations within mean.
State the inequality used in your calculation.
- d). Calculate the exact probability of part (iii) if $X \sim \text{Uniform}(0,10)$.

Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k+1, k+2, \dots$
Geometric (θ)	$P(X = x) = \theta(1 - \theta)^{x-1}; x = 1, 2, \dots$
Hypergeometric (n, N, M)	$P(X = x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$